Consider the following system of linear constraints:

$$
\begin{aligned}
x_1 + x_2 + x_3 + x_4 &= 1, \\
x_1 - x_3 &= -1, \\
2x_1 + x_2 + x_4 &= 0.
\end{aligned}
$$

Propose a procedure to eliminate as many variables as possible, except x_1. Identify the basic and non basic variables, and write explicitly the equation

$$
Ax = AP(P^T x) = Bx_B + Nx_N = b,
$$

including the matrix P.